

Nihar Gaddam

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EDUCATION

Carnegie Mellon University

Master of Science in Computer Engineering – Applied Advanced Study

Pittsburgh, PA

Dec 2023

- GPA: 3.75/4.00
- Relevant Coursework: Distributed Systems, Cloud Computing, Foundations of Computer Systems, Computer Vision, Machine Learning for Signal Processing, Introduction to Deep Learning

Sreenidhi Institute of Science and Technology

Bachelor of Technology in Electronics and Communication Engineering

Hyderabad, India

June 2020

- GPA: 9.39/10.00
- Minor in Computer Science and Engineering

SKILLS

Languages: Go, Rust, Python, C++, Java

IAAS: AWS, Azure, GCP

Container Platform: Docker, Kubernetes

CI/CD: GitHub Actions, Apache Airflow

IAC & Config Management: Terraform, Puppet

Storage: HBase, HDFS, Zookeeper, MySQL, MongoDB, Neo4j, Postgres

Frameworks/Tools: PyTorch, Spark, Kafka, Vertex AI

Monitoring and Observability: Prometheus, Grafana, Splunk, OpenTelemetry, SignalFx

EXPERIENCE

Cisco Systems Inc.

Software Development Engineer

Pittsburgh, PA

Aug 2024 – Present

Backend Engineering & API Development

- Designed and maintained features for a Go-based Kubernetes operator (control plane) managing the full lifecycle of thousands of hosted Splunk Cloud deployments — the isolated, multi-tenant product instances used by customers and internal teams — plus a customer-facing configuration API.
- Extended the customer-facing configuration API with IPv6 support and contributed to its CLI, broadening supported addressing and product capability for customers.

System Monitoring & Performance Optimization

- Held 24/7 on-call across 5 Kubernetes clusters and 8 services plus the control plane, triaging and resolving production incidents on page while reducing technical debt through targeted refactors across development, staging, and production.
- Partnered directly with customers to reproduce and resolve production-level issues in both the control plane and the configuration API.

Key Projects and Achievements

- Delivered a declarative, self-serve workflow to replace months-long manual provisioning of preview deployments (short-lived demo instances teams trial before adopting the official version), with validations that strictly isolate preview from production; scaled to ~100/month with zero manual coordination, accelerated customer testing workflows 38%, and drove adoption across five teams.
- Built the platform's first metrics + SLO observability layer (Prometheus, OpenTelemetry), instrumenting per-phase latency and success/failure metrics across every provisioning stage — infrastructure, software, and final delivery — with data-driven SLOs; shipped pageable alerts that catch genuine regressions proactively and cut cross-team mispages by 21%. Extended it with a dedicated OpenTelemetry collector sidecar for the provisioning operator to cover teardown as well as provisioning, and corrected phase-specific histogram buckets to eliminate inflated percentile estimates while holding down metric-stream cost.
- Diagnosed and fixed a disaster-recovery teardown ordering bug that could destroy a live primary deployment while its replica was still active; redesigned the control loop to enforce dependency-safe teardown, preventing data loss and eliminating the manual-cleanup incident tickets and on-call pages those mis-ordered teardowns had generated.
- Built an automated garbage-collection system for stale infrastructure, reclaiming 2,385 abandoned deployments and cutting unnecessary cloud spend fleet-wide in production alone.
- Automated service-to-service authentication with ephemeral credentials, removing the need to raise manual tickets to update the latest build version, and drove encryption-at-rest to 100% coverage across 3,000+ deployments.
- Led a backward-compatible migration from PBKDF2 to HKDF (AES-GCM) across the platform's cryptography library and the Kubernetes control plane, adding a versioned cipher codec, CLI and API support, automatic codec detection, and certificate-encryption contracts behind a version-gated rollout; verified bidirectional compatibility on staging deployments so existing and newly written ciphertext stay readable fleet-wide.

- Authored a design proposal integrating a downstream pooling service with the control plane to pre-provision Azure subscriptions for deployment networking (otherwise created on demand), raising Azure network provisioning success from 33% to 98% and cutting provisioning time to ~20 min.
- Automated authentication-token rotation across 15 service accounts, eliminating manual intervention and improving token-management efficiency ~40%.

Accenture Solutions Pvt. Ltd.

Application Development Associate

Hyderabad, India

Dec 2020 – Oct 2021

Cloud Infrastructure Management

- Migrated client infrastructure from on-premises to AWS and Azure, improving scalability and reducing operational costs by 20%; maintained development and production environments, reducing downtime by up to 40%.
- Executed high-risk production changes with minimal disruption, maintaining a 99.8% success rate in change management.

Automation and Scripting

- Automated collection of performance, CPU, and disk-usage data across development and production servers with scheduled PowerShell jobs, replacing manual per-server checks.
- Built an automated workflow to retrieve, cleanse, and organize data from Snowflake into Azure SQL, keeping datasets development-ready without manual preparation.

Data Analysis and Reporting

- Leveraged Power BI for real-time server performance trend analysis, reducing response time to performance issues by 30%; managed ETL dataflows from Azure SQL, SharePoint, Redshift, MongoDB, and Snowflake into BI reporting pipelines.
- Cleaned and transformed raw Excel and Oracle data with Power Query, automating data-integration steps and improving data accuracy 22%.
- Stood up and maintained a Power BI data gateway bridging on-premises databases (Oracle, MySQL, PostgreSQL) to the reporting layer.
- Built an error-logging system in Power BI with custom error-log tables and automated capture and notification, enabling quick identification of data-refresh failures and reducing reporting downtime.

PROJECTS

timberdb | Go

Jun 2026

- Built a time-partitioned, TTL-native LSM storage engine in Go; benchmarked 2.0x Pebble, 5.8x Badger, and 13x bbolt append throughput at 3 allocs/op, with 6.6x faster scans than Badger.

SVID-Exchange | Go

Mar 2026

- Built a Zero Trust token exchange service that converts SPIFFE SVIDs into scoped, short-lived ES256 JWTs over mTLS, eliminating static shared secrets for service-to-service authentication.

Saga-Conductor | Go

Mar 2026

- Built a lightweight saga-pattern orchestrator for distributed transactions: callers define each step with a forward action and a compensating action, and the conductor executes them in order, automatically rolling back completed steps in reverse on failure.

CERTIFICATIONS

Certified Kubernetes Application Developer (CKAD)  — Cloud Native Computing Foundation (CNCF) 2024